

Tracking Employment Changes in AI-Exposed Jobs

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Abstract

This paper examines labor market trends for occupations exposed to Generative AI using US Current Population Survey (CPS) data from the post-ChatGPT era (Q4 2022 - Q1 2025). Contrary to displacement fears, occupations most exposed to AI on average show no substantial difference in employment or earnings growth compared to the least exposed. However, this aggregate trend masks heterogeneity across occupations. Within the most exposed quartile, occupations with a higher share of college-educated workers, such as software development, have experienced robust employment growth, while those with a lower share, like customer service, have seen declines. The analysis also reveals a divergence between job posting data and realized employment, urging caution when inferring labor market trends from postings alone.

1 Introduction

The period following the public release of ChatGPT in November 2022 has seen burgeoning research attempting to capture the labor market and productivity effects of generative AI, spurred by fear of potential widespread job displacement. This paper provides a timely empirical analysis of this evolving landscape by leveraging U.S. Current Population Survey

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(CPS) data through Q1 2025. Using the GPT-4-based occupational exposure metrics from [Eloundou et al. \(2024\)](#), I examine realized employment and earnings trends to measure whether, and how, disruption from new AI technologies is appearing in the U.S. workforce.

The central finding is that the most AI-exposed set of occupations has not experienced a declining share of employment, though among this group there is considerable heterogeneity in labor market trends. Software occupations, which are estimated to have high exposure to generative AI, show employment gains as of the first quarter of 2025. In contrast, customer service representatives, who also have high measured exposure, have experienced relative employment declines. More broadly, within the most exposed quartile, occupations with a higher share of college-educated workers have experienced faster employment growth over the prior four years compared to occupations with a lower college share. The results suggest a nuanced view of how measured AI or automation exposure affect employment, with worker characteristics such as education potentially influencing displacement from new technologies. While the results do not identify a causal impact of AI on labor market outcomes, they clarify facts about the state of the US labor market and suggest avenues for further study on how AI impacts workers.¹

Given growing concern over AI’s impact on early-career roles, I also examine trends for entry-level workers ([Thompson, 2025](#); [Raman, 2025](#); [Allen, 2025](#)). Employment for entry-level workers is volatile and cyclical, making it difficult to discern trends. With more time and larger surveys it may become apparent whether there are sustained changes in labor market outcomes for entry-level workers, but so far evidence from the CPS is inconclusive.

A further contribution of this paper is to highlight the divergence between employment and job postings. While Indeed job postings for software development show a decline from a peak in early 2022, CPS employment data indicate stable or even increasing employment through Q1 2025. The disconnect between job postings and employment also tracks with

¹Another way to interpret the results is that current AI exposure and automation measures may do a poor job of predicting the true labor market impacts of AI, at least in the relatively short time period considered here. This contrasts with findings in prior seminal work such as [Autor et al. \(2003\)](#) that showed that task exposure to computerization strongly predicted changes in relative labor demand.

a relatively stable US national unemployment rate and labor force participation rate over the same period. These findings urge caution in interpreting labor market trends from job postings alone.

Because the CPS is released monthly and is publicly accessible, the method in this paper can be used to track changes in occupational employment on an ongoing and transparent basis. While the CPS’s sample size limits statistical power for granular groups like entry-level workers, its strength lies in providing a consistent, high-frequency view of the entire U.S. labor market. This paper thus clarifies the current state of facts and provides a foundation for future work using larger-scale administrative or internal Census data to further probe the mechanisms behind these divergent outcomes.²

2 Related Literature

This paper engages with three active areas of research on the labor market effects of artificial intelligence. First, it builds on literature that measures AI’s *potential* exposure. A series of influential papers established methodologies for estimating which occupations were susceptible to automation (Frey and Osborne, 2017; Brynjolfsson et al., 2018; Felten et al., 2018, 2019; Webb, 2019; Felten et al., 2021). More recently, work such as Eloundou et al. (2024); Felten et al. (2023); Gmyrek et al. (2023), and Gmyrek et al. (2025) adapted this approach for Generative AI, forming the basis for exposure metrics used in this analysis. While these studies identify potential disruption, this paper’s contribution is to use broad, representative U.S. survey data (the CPS) to provide one of the earliest empirical snapshots of realized employment changes in the post-ChatGPT era. This complements studies that find significant effects but in more specific settings, such as on online freelance platforms (Hui et al., 2023) or within individual firms (Brynjolfsson et al., 2025; Dillon et al., 2025).³

Second, this paper contributes to the growing body of work using economy-wide data to

²All code is posted online at https://github.com/chandarb/CPS_tracker.

³See also Noy and Zhang (2023); Peng et al. (2023); Dell’Acqua et al. (2023).

measure AI’s impact. Recent findings have been varied. [Humlum and Vestergaard \(2025\)](#) use Danish administrative data to find minimal effects on earnings or hours worked, while [Jiang et al. \(2025\)](#) find AI exposure is correlated with longer work hours in the U.S.⁴ This paper adds additional color to this growing set of findings. In line with [Humlum and Vestergaard \(2025\)](#), I find little evidence of an aggregate decline in employment for exposed occupations. However, in contrast to their work, I find heterogeneity in employment trends across exposed occupations, with these differences correlated with features such as education. The divergent paths of software developers and customer service representatives, for example, suggest nuance in predictions of labor market changes for exposed workers.⁵

Finally, this work highlights an important methodological issue: the divergence between job postings and employment data. Recent studies have used job postings to measure shifts in labor demand from AI (e.g., [Acemoglu et al. \(2022\)](#); [Hampole et al. \(2025\)](#)). This paper shows that for key occupations, these two data sources tell different stories. The decline in software developer job postings since 2022, for instance, contrasts sharply with their stable-to-growing employment in the CPS. This finding serves as an important empirical caution in interpreting job posting trends as reflective of employment or earnings. The CPS data used here provides a transparent and timely alternative for tracking realized outcomes as they unfold for a representative sample of the U.S. population.

⁴See also [Acemoglu et al. \(2022\)](#); [Bonney et al. \(2024\)](#); [Bick et al. \(2024\)](#); [Hartley et al. \(2024\)](#); [Frank et al. \(2025\)](#); [Chen et al. \(2025\)](#); [Hampole et al. \(2025\)](#). The Federal Reserve Bank of New York tracks employment by college major using the American Community Survey (ACS), as documented in [Abel and Deitz \(2017\)](#). While the ACS has rich sociodemographic and economic information, it is released annually and with a time lag. As of Q2 2025 the latest ACS data is from 2023, in the initial year of LLM adoption.

⁵Another closely related paper is [Deming et al. \(2025\)](#), which combines data from the decennial Census, American Community Survey, and CPS to track labor market changes over 120 years. This paper complements their work by directly associating employment changes with measures of AI exposure and by tracking labor market trends specifically over the post-ChatGPT time period.